

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2019

BIOLOGY PAPER 1

SECTION B : Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.
- (2) Refer to the general instructions on the cover of the Question Paper for Section A.
- (3) Answer **ALL** questions.
- (4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.
- (6) Present your answers in paragraphs wherever appropriate.
- (7) The diagrams in this section are **NOT** necessarily drawn to scale.
- (8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

Please stick the barcode label here.

Candidate Number

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SECTION B

Answer **ALL** questions. Write your answers in the spaces provided.

1. (a) Physical and chemical barriers are the first line of defence in the human body. Select from Column II **all** correct example(s) that belong(s) to the two types of barriers in Column I and put the letter(s) in the spaces provided. (2 marks)

Column I

Column II

(i) physical barrier _____

A. skin

B. tear

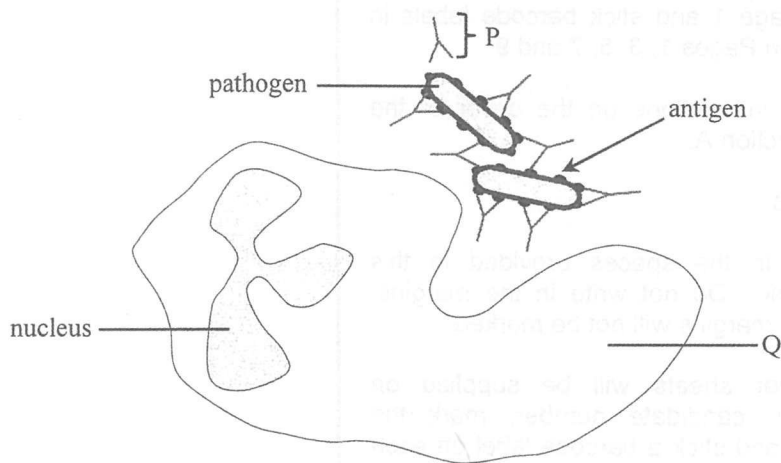
(ii) chemical barrier _____

C. antibody

D. blood clot

E. gastric juice

- (b) The diagram below shows the process of phagocytosis. Q is a phagocyte while P is a protein molecule produced by a type of lymphocyte.



Describe the function of P in phagocytosis.

(3 marks)

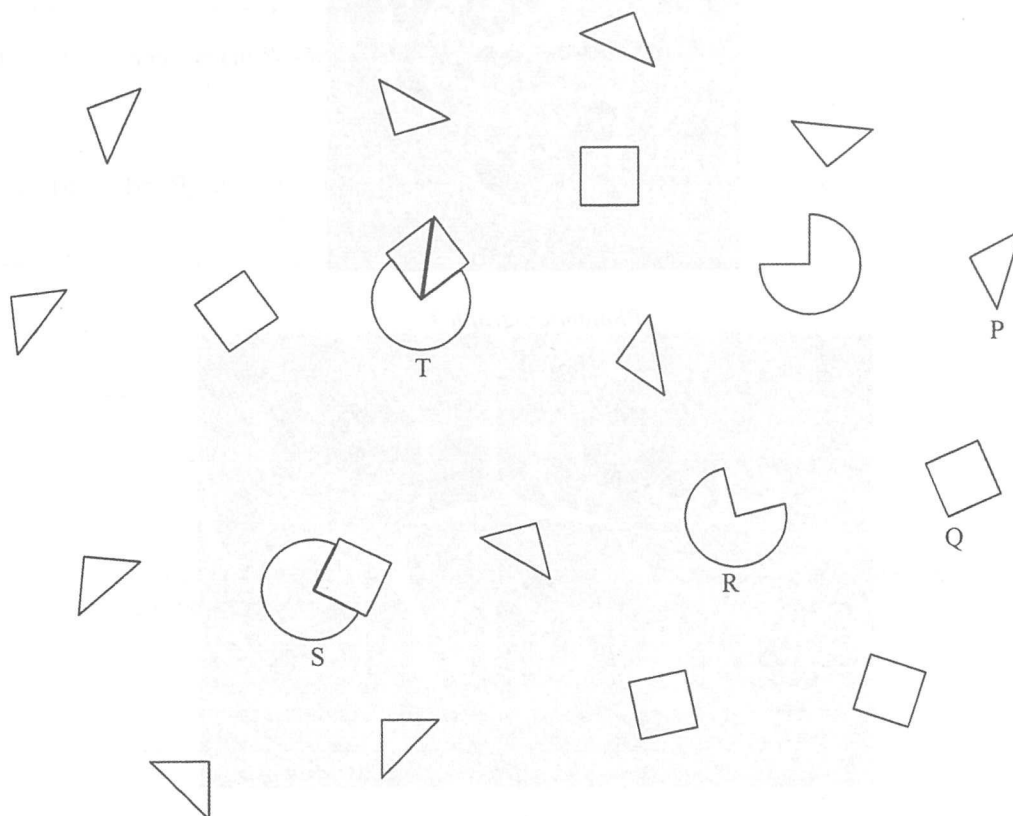
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2. The schematic diagram below shows a reaction mixture of an anabolic reaction catalysed by an enzyme. Drawings P, Q, R, S, and T represent different components of the mixture:



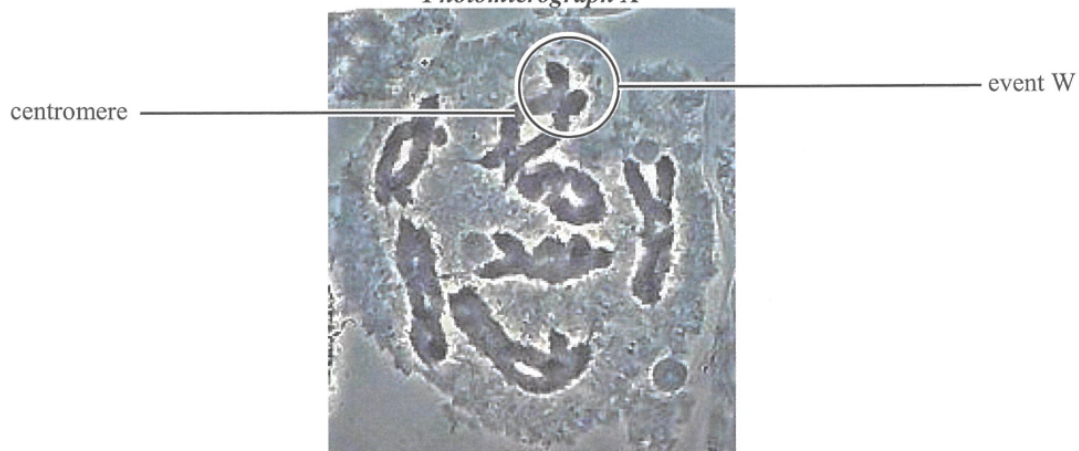
- (a) Which drawing represents the substrate in this anabolic reaction? Explain your answer. (2 marks)

- (b) Which drawing represents the enzyme? Explain your answer. (2 marks)

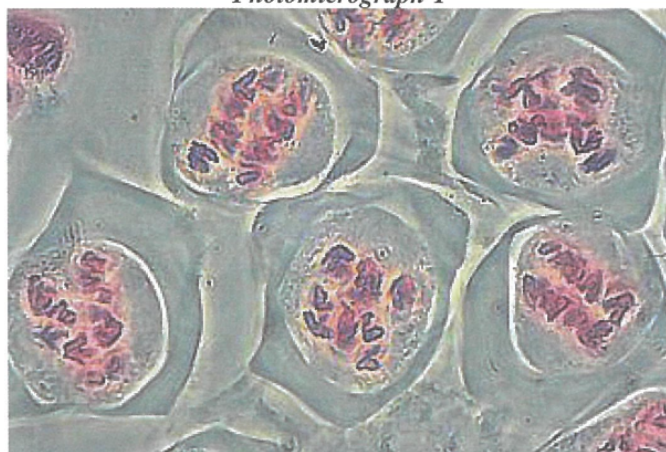
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3. The photomicrographs below show some stages of meiosis taking place in a flower:

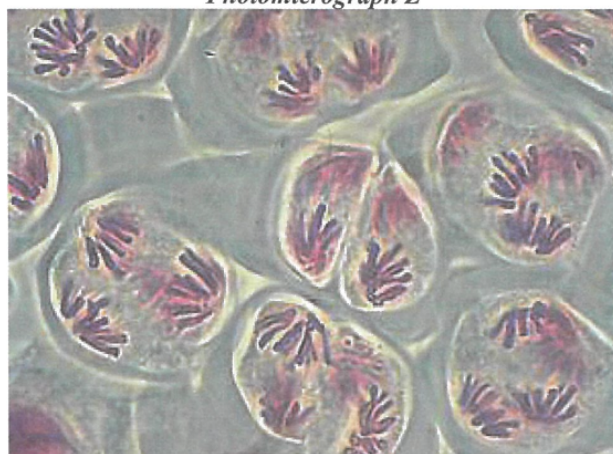
Photomicrograph X



Photomicrograph Y



Photomicrograph Z



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Please stick the barcode label here.

- (a) State *one* floral structure in which this type of division takes place. (1 mark)

- (b) (i) Name event W shown in Photomicrograph X. (1 mark)

- (ii) Briefly describe what happens in event W. What is the importance of event W? (2 marks)

- (c) (i) Which photomicrograph, Y or Z, shows the first meiotic division? Give a piece of evidence to support your answer. (2 marks)

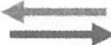
- (ii) What is the purpose of the first and second meiotic divisions respectively? (2 marks)

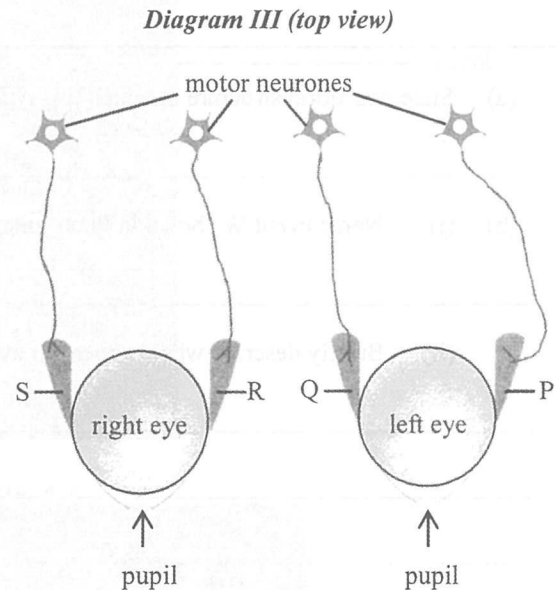
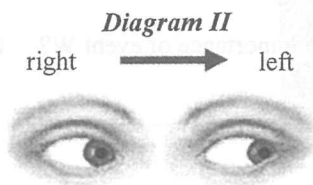
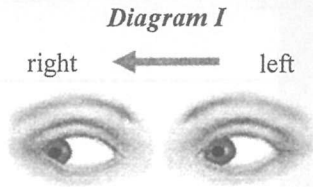
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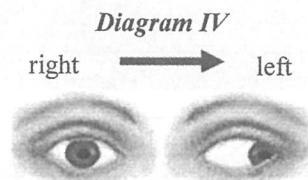
4. Diagrams I and II below show a person with both eyes moving right and then left. This eyeball movement is brought about by the coordination of different eye muscle pairs. Diagram III shows four of the muscles (P, Q, R, and S), all connected to motor neurones controlling eyeball movement.

Key:  directions of eyeball movement



- (a) To bring about the eyeball movement shown in Diagram I, which muscle(s) (P, Q, R, or S) contract(s)? (1 mark)

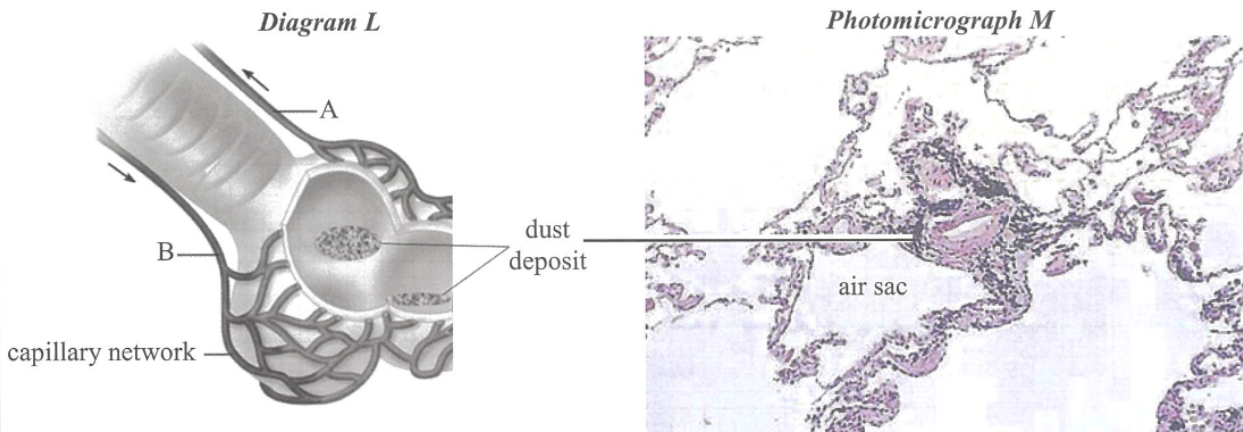
- (b) A person suffers from impaired eyeball movement when turning his eyes from right to left, as shown in Diagram IV.



It is found that one of his eye muscles cannot fully contract. Based on your knowledge of neurotransmission at the neuromuscular junction, suggest *two* possible defects that would lead to the impaired eyeball movement shown in Diagram IV. (2 marks)

Answers written in the margins will not be marked.

5. Diagram L below shows part of the lung in a patient suffering from a certain lung disease. A hardened layer of dust deposit was found on the respiratory surface of the air sacs. Photomicrograph M shows the lung tissue taken from the patient.



- (a) Compare the oxygen and glucose content of the blood in vessels A and B. Explain your answer. (4 marks)

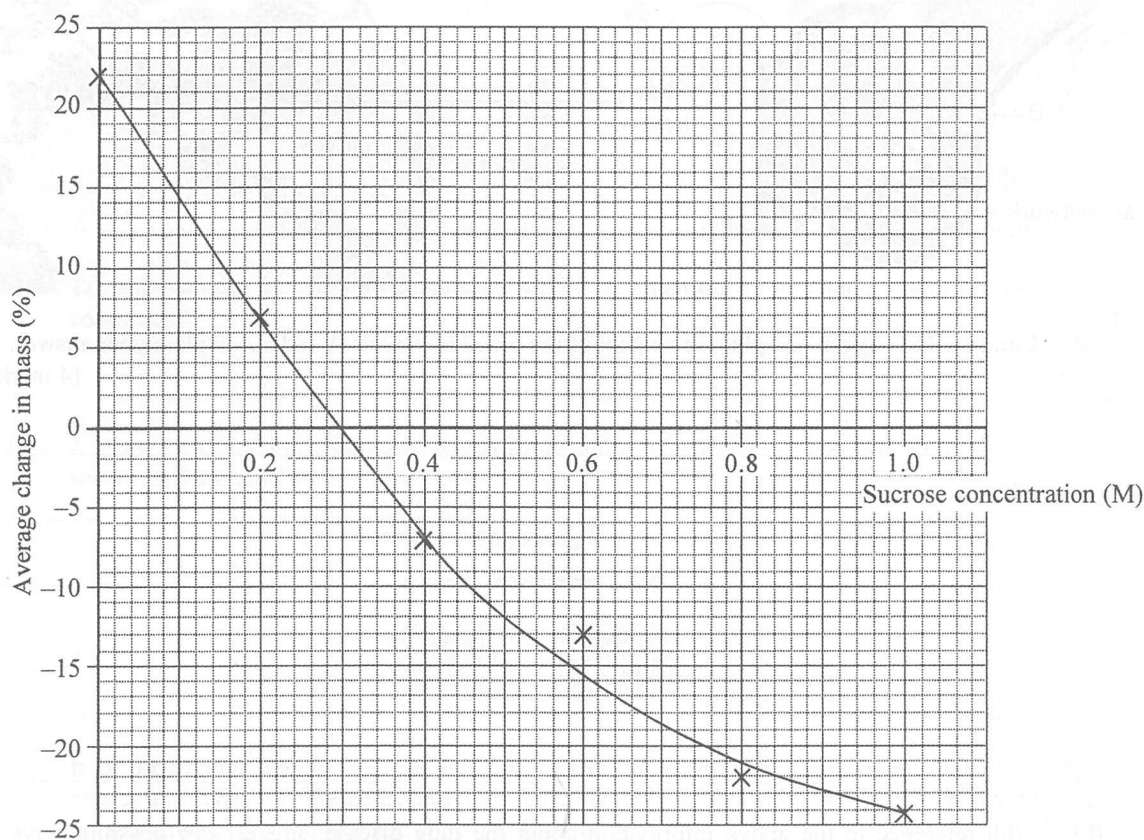
- (b) With reference to the above information about the lung disease, suggest *two* possible ways in which the disease adversely affects gas exchange in the patient. (4 marks)

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6. Johnny conducted an experiment to determine the water potential of potato tuber cells. He measured the masses of fresh potato cylinders before and after immersing them in sucrose solutions at different concentrations (0M, 0.2M, 0.4M, 0.6M, 0.8M, and 1.0M). Below shows the major steps in the experiment:

- Step 1: Cut potato tubers into cylinders
- Step 2: Blot dry the surface of the potato cylinders
- Step 3: Weigh the potato cylinders (initial mass)
- Step 4: Immerse three potato cylinders in each concentration of sucrose solution for two hours
- Step 5: Remove and blot dry the surface of the potato cylinders
- Step 6: Reweigh the potato cylinders (final mass)
- Step 7: Calculate the average percentage change in mass of the potato cylinders in each solution

The results are shown in the graph below:

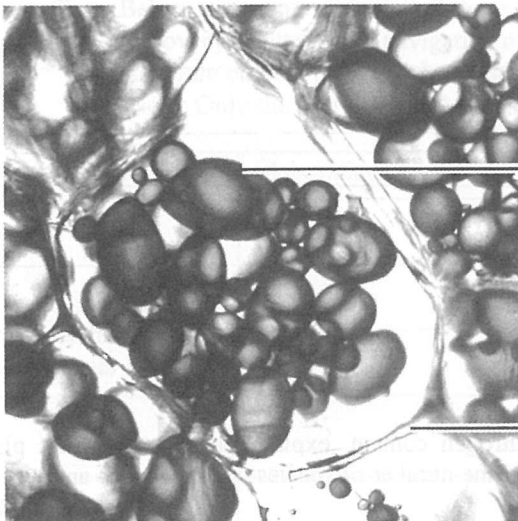


- (a) With reference to the graph, which sucrose solution concentration has the same water potential as the potato cells? Explain your answer. (3 marks)

Answers written in the margins will not be marked.

- (b) If Johnny skipped step 2 by mistake for all samples, how would this affect the curve and the deduced value of the concentration of the sucrose solution in (a)? Sketch a curve on the graph on the facing page to show the effect. (1 mark)
- (c) In terms of experimental design, what is the importance of putting three potato cylinders in each concentration of sucrose solution? (1 mark)

- (d) Johnny prepared a slide of freshly sectioned potato cylinder and stained it with iodine solution. The photomicrograph below shows the section. Label structures X and Y. (2 marks)



X: _____

Y: _____

- (e) In the middle of the 19th century, there was a severe attack on potato crops by a plant pathogen in Ireland. As the potato was the major staple food at that time, many Irish people died of starvation. Vegetative propagation of potatoes was blamed for the high vulnerability of the potato crops. Explain the rationale for this claim. (2 marks)

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7. The table below shows the changes in soil nitrogen content and the number of species of herbaceous plants and woody plants before and after a landslide on a hillside:

	Soil nitrogen content (mg g ⁻¹)	Number of plant species	
		Herbaceous plants (e.g. grass)	Woody plants (e.g. shrubs and trees)
Before landslide	6	10	15
2 years after landslide	1	17	2
20 years after landslide	3	14	9

- (a) What type of ecological succession occurred on the hillside after the landslide? Explain your answer. (2 marks)

- (b) (i) Explain the change in the soil nitrogen content shown in the above table. (3 marks)

- (ii) With reference to the change in soil nitrogen content, explain the change in the plant composition after the landslide in terms of the number of species of herbaceous and woody plants. (3 marks)

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8. (a) Bats are night-time animals. The mystery of how bats avoid obstacles in darkness has puzzled scientists for centuries. Below shows some major events in the research on how bats use ultrasound to navigate as they fly:

Time	Scientist	Event
Late 18 th Century	Spallanzani	He noted that blind bats could avoid obstacles.
	Jurine	He plugged the ears of bats with wax; the bats collided with obstacles.
Next 140 years	Various scientists	Despite the work of Spallanzani and Jurine, scientists continued to explore the possible use of other senses for navigation in bats.
1930s	Pierce	He developed an apparatus that could detect ultrasound.
1938	Griffin	He used Pierce's apparatus to show that bats emitted ultrasound.
	Griffin and Galambos	They worked out how bats used the ultrasound they produced in navigation.

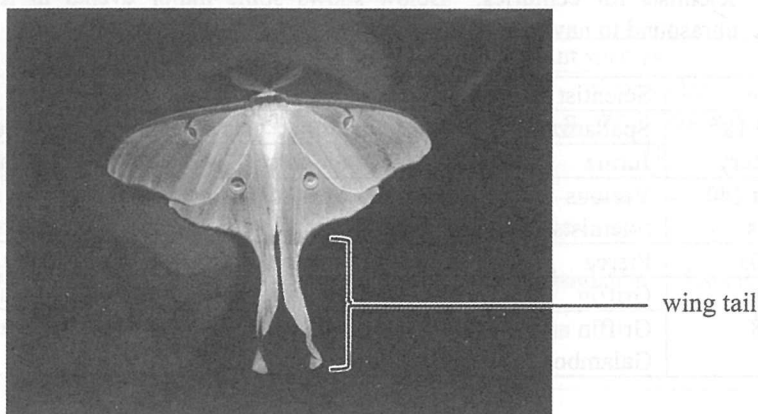
- (i) With regard to the observation of Spallanzani, what conclusion can you draw about bats' ability to avoid obstacles? (1 mark)

- (ii) Below are some aspects of the nature of science which can be demonstrated by the discovery of ultrasound navigation of bats. Choose any *two* aspects of nature of science and elaborate on how these are demonstrated in the above events: (2 marks)
(Note: Only the first two will be marked if you give more than two aspects.)

Nature of Science	Elaboration
Science is based on evidence from experiments	
Scientists build on the work of other scientists	
Technology has impacts on the development of science	

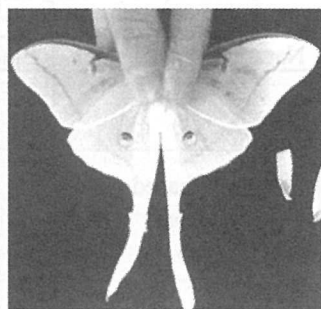
Answers written in the margins will not be marked.

- (b) Bats prey on moths. The photograph below shows a type of moth which has long wing tails:



After the discovery of ultrasound navigation in bats, scientists hypothesised that the wing tails of the moths may disturb the ultrasound emitted by bats and thus help moths to escape from a bat attack.

To test this hypothesis, scientists manipulated the wing tail length of the moths and then determined their rate of successful escape from bat attacks. The treatments of the wing tails and the results are shown below:

Treatment of wing tails	A: No treatment	B: Cut and glued back	C: Cut	D: Elongated
				
Wing tail length (cm)	5	5	2	6
Successful rate of escape (%)	57	57	26	65

- (i) What can you conclude from the results of treatment A and B? What is the purpose of treatment B in the experimental design of this study? (2 marks)

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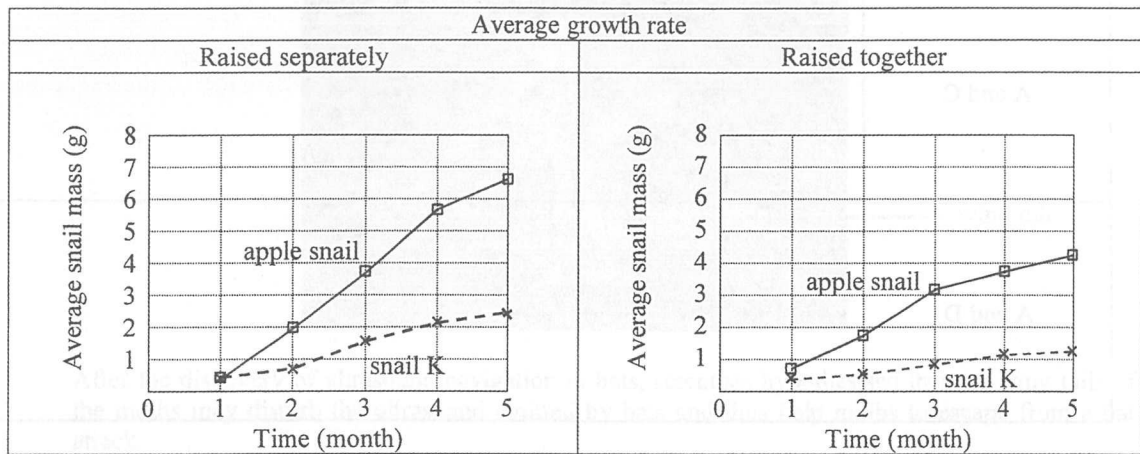
- (ii) What further conclusion can you draw when comparing the results of the following treatments? (2 marks)

Treatment	Conclusion
A and C	
A and D	

- (iii) What is the overall conclusion of this study? (1 mark)

- (c) With reference to the hypothesis stated in (b), describe how the long wing tail could have evolved in the moths. (4 marks)

9. The apple snail originates from South Africa. It was first imported to Asian countries for human consumption. However, it escaped to the local wetland habitat. Below are data regarding the average growth rates of apple snail and a local snail species K when they were raised separately and raised together:



- (a) With reference to the above data, what would happen to the population of snail K in the wetland habitat once the apple snails have escaped to this habitat? Support your answer with data from the graphs above. (4 marks)

- (b) It has been noted that apple snails consume wetland plants at a high rate, especially buds and young leaves. Suggest why the feeding habits of apple snails may have an adverse effect on the community of local wetland habitats. (3 marks)

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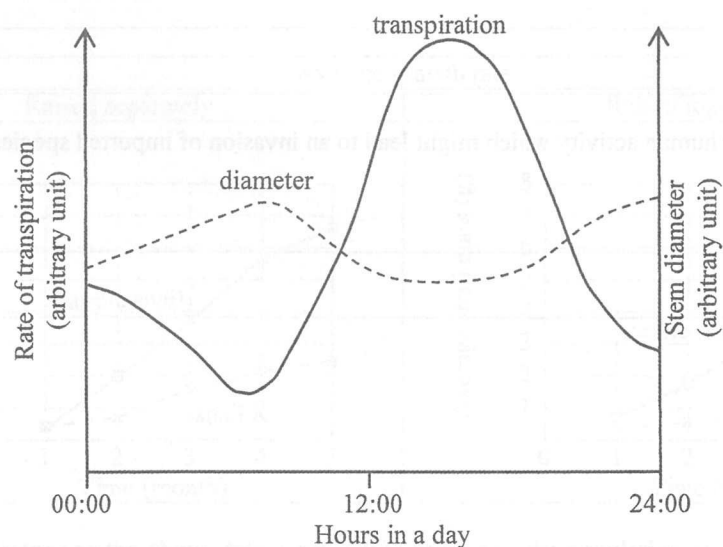
- (c) Apart from the above, suggest another biotic factor which may explain why an imported species would turn into an invasive or dominant species. (1 mark)

- (d) Suggest *one* human activity which might lead to an invasion of imported species in Hong Kong. (1 mark)

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10. The graph below shows the change in the rate of transpiration and the change in stem diameter of a plant over 24 hours:



- (a) Describe the relationship between the rate of transpiration and stem diameter. (1 mark)

- (b) It is known that the change in stem diameter is related to the diameter of the xylem vessels. With reference to the way in which water is transported along the stem, explain the relationship between the rate of transpiration and stem diameter described in (a). (2 marks)

- (c) Describe and explain *two* adaptive features of xylem vessels as a structure for water transport. (4 marks)

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You are required to present your answer in essay form to the following question. Criteria for marking will include relevant content, logical presentation and clarity of expression.

11. Nowadays, keeping pets (such as dogs and cats) is becoming popular in Hong Kong. Some people prefer pure-bred pets to hybrid pets. However, pure-bred pets usually have higher risks of suffering from genetic diseases than hybrid pets because of the ways they are bred. Pure-bred pets are produced by crossing close relatives to keep a pure bloodline. Explain why genetic diseases are often carried by recessive alleles. By comparing the effects of the two breeding processes on the genetic composition of the offspring, discuss why pure-bred pets are at a higher risk of suffering from genetic diseases than hybrid pets. (11 marks)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

Answers written in the margins will not be marked.